Abdur Rahman

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I am a curious and driven learner with a strong passion for coding, analytical problem solving and continuous self improvement. I thrive in collaborative environments where I can both contribute and learn from others and take initiative to tackle challenges. Known for my dedication and discipline, I enjoy leading through actions, supporting my peers and finding efficient and creative solutions to complex technical problems. Currently exploring AI/ML and want to do impactful work using it.

Education

Bachelor of Science in Computer Science and Engineering CGPA: 3.93/4.00 *2023 – Present*

American International University–Bangladesh (AIUB), 408/1 (Old KA 66/1), Kuratoli, Khilkhet, Dhaka 1229, Bangladesh

- **Relevant Coursework:** Data Structures, Algorithms, Object Oriented Programming I & II, Object Oriented Analysis and Design, Introduction to Database, Software Engineering, Artificial Intelligence and Expert System, Web Technologies, Computational Statistics and Probability, Machine Learning, Introduction to Data Science, Computer Vision and Pattern Recognition, and Natural Language Processing.

Anticipated Graduation: June/2027
Current Progress: Completed 9th semester and 127 of 148 credits.

Projects

End-to-End Water Quality Classifier *github.com/End-to-End-water-quality-Classifier*

Machine Learning, Python, Scikit-learn, Gradio

- Developed an end-to-end ML pipeline for water potability prediction using Random Forest, incorporating missing value imputation, feature scaling, GridSearchCV and cross validation.
- Built an interactive Gradio web interface for real time predictions, evaluated model performance using classification metrics and confusion matrix analysis and serialized the trained pipeline using Pickle.

Fashion-MNIST Multiclass Image Classification using CNN *github.com/Fashion-mnist-CNN-pytorch*

Deep Learning, PyTorch

- Designed a four block CNN in PyTorch for Fashion-MNIST classification, using batch normalization, pooling and dropout for feature extraction and regularization.
- Processed grayscale image data into tensor representations and trained the end-to-end deep learning pipeline, achieving 90.65% test accuracy across 10 clothing categories.

Simple GK Answering System using RNN *github.com/Simple-gk-answring-system-using-rnn*

Natural Language Processing, PyTorch, RNN

- Developed a General Knowledge question-answering system using an RNN in PyTorch, with text cleaning, tokenization, vocabulary creation, numerical encoding and word embeddings.
- Built a custom PyTorch Dataset and DataLoader and trained an Embedding RNN Linear architecture using CrossEntropyLoss and the Adam optimizer for answer token prediction.

LSTM Next Word Predictor *github.com/lstm-next-word-predictor-pytorch*

Natural Language Processing, PyTorch, LSTM, NLTK

- Developed an LSTM based next word predictor in PyTorch, implementing NLTK tokenization, vocabulary construction, sequence generation, padding and custom data loading.

- Trained an Embedding LSTM Linear model using CrossEntropyLoss and Adam, achieving 96.94% training accuracy and generating text autoregressively with greedy decoding.

Human Resource Management System *github.com/Human-Resource-Management-System*

Object Oriented Programming II, AIUB (C#)

- Developed a .NET HR management application supporting user, employee and employer workflows with file/database storage for secure and persistent employee records.
- Applied core OOP concepts including abstraction, inheritance and event driven programming in C# to build and manage routine HR functionalities.

E-Commerce Store Management System *github.com/E-Commerce-Store*

Web Technologies, AIUB (PHP, MySQL, AJAX, JavaScript, HTML, CSS)

- Developed a full stack e-commerce platform using PHP and MySQL, featuring product CRUD operations, category management, image uploads, stock tracking and database integrated inventory management.
- Implemented admin side functionality including category hierarchy handling, AJAX-based live availability toggling, responsive product management and real time stock alerts.

Interactive 2D Animated Environment using OpenGL *github.com/Computer-Graphics*

Computer Graphics, AIUB (C++, OpenGL, GLUT)

- Developed an animated 2D environment using OpenGL and GLUT, featuring dynamic weather, day-night transitions, sports simulations, carnival scenes and real time object interactions.
- Applied computer graphics techniques including translation, rotation, scaling, alpha blending, dynamic lighting, animation loops and coordinate based rendering for interactive scenes.

Shop Management System *Spring 2023-2024*

Object Oriented Programming I, AIUB (Java)

- Designed and implemented a desktop application for small retail operations, including inventory, billing and user management.
- Applied OOP principles (encapsulation, inheritance, polymorphism); used file I/O for data persistence.

Technical Strengths

Programming Languages: C++, C, Python, C#, MySQL, Java, Java Script, PHP, HTML/CSS
HDLs: VHDL (Basic)
Developer Tools/Frameworks/Libraries: Notepad++, Git, GitHub, Pandas, NumPy, Matolotlib, Seaborn, scikit-learn, Google Colab, Gradio, Visual Studio, Jupyter Notebooks, PyTorch, Git, VS Code
Others: LaTeX, MATLAB, Proteus, NI Multisim, Notion, Trello
Languages: English, Bangla (Native & Fluent)

Scholastic Achievements & Awards

- **Dean's List Honors** (AIUB) in 4 semesters for achieving GPA 3.90, 3.91 and 4.00(2).
- Selected for International Collegiate Programming Contest **(ICPC)** Asia Dhaka Regional on behalf of AIUB in 2024.
- **3rd place** in School Science Fair, 2015 (Developed an eco-friendly electricity generation concept).
- Merit Scholarship based on AIUB academic & previous results **(Up to 60% tuition waiver)**.
- Successfully completed the **"Intermediate Importing Data in Python"** certification course from DataCamp.
- Successfully completed the **"Preprocessing for Machine Learning in Python"** certification course from DataCamp.

- Successfully completed the **"IT Essentials"** course from Cisco Networking Academy.

Seminar & Conference

7th Bangladeshi Industrial Engineering and Operations Management International Conference *December 2024*

Title: A Statistical Analysis on Unemployment After Graduation in Bangladesh

Presented a poster on unemployment among Bangladeshi graduates using Secondary data set and statistical analysis.

Workshop on Persuasive Presentation *(AIUB English Club)* *August 2025*

Enhanced public speaking, persuasive presentation, communication and audience engagement skills.

LinkedIn Profile Optimization Workshop *(AIUB English Club)* *November 2025*

Gained practical knowledge of LinkedIn optimization, personal branding, professional networking and communication.

Practical DevOps with AWS Workshop *(Cloudly Infotech Limited & AWS Partner)* *April 2026*

Gained hands on exposure to AWS cloud services, DevOps practices, cloud infrastructure and deployment workflows.

Extracurriculars & Skills

Associate Member, United Nations Youth and Students Association of Bangladesh (UNYSAB) *Present*

Member, AIUB English Club *Spring 2024 – Present*

Actively participated in events, debates and speaking sessions, strengthening communication, public speaking, teamwork, leadership and confidence through collaborative activities.

Problem Solving *September 2023 – Present*

Solved 1000+ problems using C++. Primary platforms profiles: CodeForces , LeetCode

Entrepreneurship *November 2020 – June 2021*

Started a phone-cover business at age 16; sold via Instagram. Cover Station

Played Cricket (Club Level) *2017 – 2018*

Aswad Cricket Academy

Played Football in CLEAR MEN U17 *April 2018*

Represented school at U17 level

Drawing *2014*

First place in school drawing competition (Class 5).